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Math Behind KNN

→ lazy algorithm
→ The most popular one.

Instead of learning patterns and making predictions, kNN remembers every single detail of the training data.

→ Similar objects tend to be found near each other.

- To know the k neighbors, the algorithm measures distance using method like Euclidean & Manhattan.

Euclidean
→ Works well for continuous data.

Minkowski

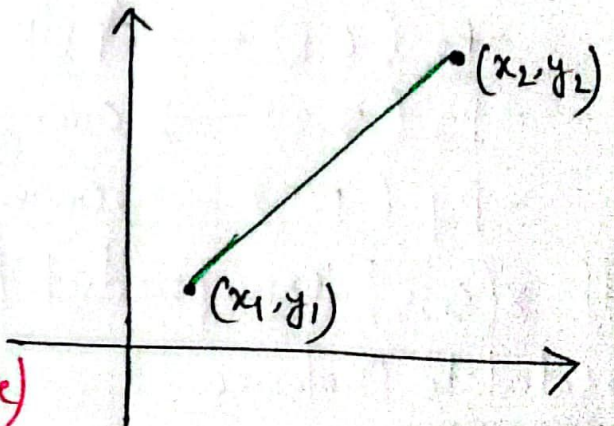
Manhattan
→ Go-to for categorical data.

Hamming Distance
→ Number of non-similar string bits.

- For classification → kNN looks for the most common class among k-neighbors.
- For regression → Calculates the average or median of the neighbors value.

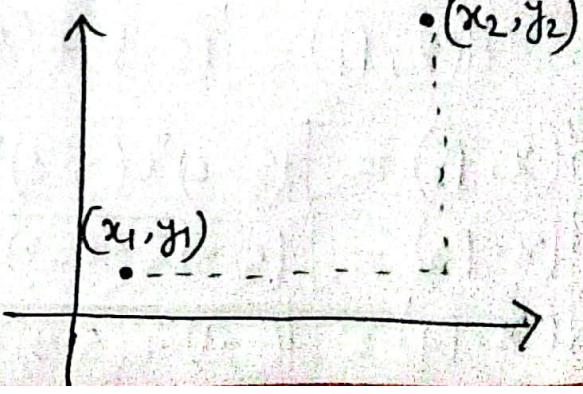
Euclidean Distance:

$$\left[d(x,y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \right]$$



Manhattan Distance: (City Block Distance)

$$\left[d(x,y) = \sum_{i=1}^{n} |x_i - y_i| \right]$$



Minkowski Distance:

$$\left[d(x,y) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p} \right]$$

- After finding the distances, the algorithm sorts & selects the smallest distance. Then the most common class label among the k- nearest neighbors is chosen as prediction.

classification

$$\left[c_q = \text{mode} \{ c_{n1}, c_{n2}, c_{n3} \cdots c_{nk} \} \right]$$

c_q → predicted class; c_{n1} → class of the 1st nearest data

Regression

$$\left[v_q = \frac{1}{k} \sum_{i=1}^{k} v_{ni} \right]$$

v_q → Predicted value; v_{ni} → target value of the k-nearest neighbors

Math: We want to classify an unknown fruit based on two features Sweetness (x_1) & Crunchiness (x_2). We choose $k=3$:

- Training Dataset: (class label)
  - P_1 (7,6) → Apple
  - P_2 (6,8) → Apple
  - P_3 (3,2) → Orange
  - P_4 (1,3) → Orange

If $x(5,5)$ then predict the fruit?

- Calculate Distances:

$$d(x, P_1) = \sqrt{(5-7)^2 + (5-6)^2} = 2.24$$

$$d(x, P_2) = \sqrt{(5-6)^2 + (5-8)^2} = 3.16$$

$$d(x, P_3) = \sqrt{(5-3)^2 + (5-2)^2} = 3.61$$

$$d(x, P_4) = \sqrt{(5-1)^2 + (5-3)^2} = 4.47$$

- Sort & Select Neighbors: (k=3)

1) P1 → Distance (2.24) → Apple
2) P2 → Distance (3.16) → Apple
3) P3 → Distance (3.61) → Orange

- Perform Voting:
  - Apple (2 Votes)
  - Orange (1 votes)

Since Apple got majority so the unknown point will indicate an apple.

